Optimizing Healthcare Operations: A No-Show Analysis

Report

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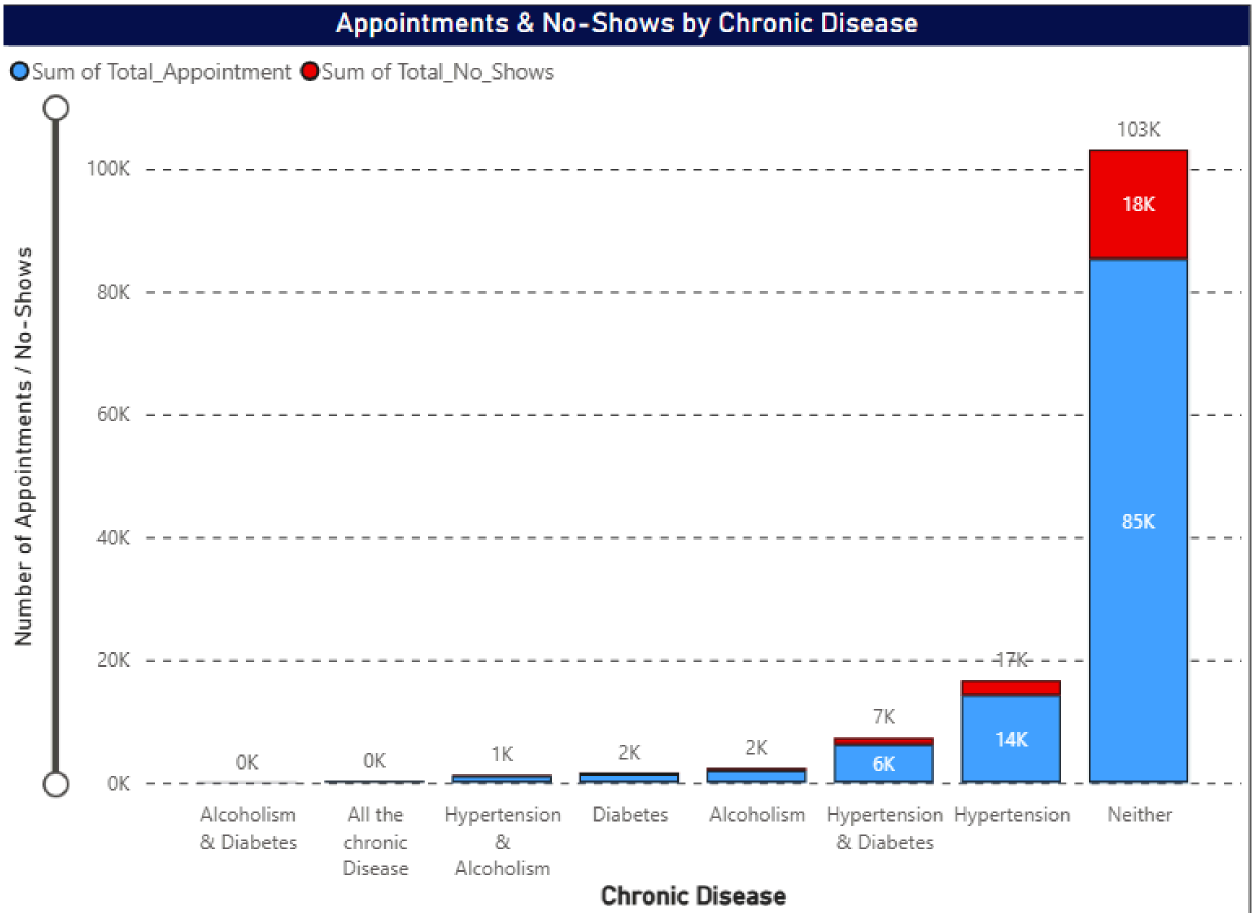
Executive Summary

The Patient Appointment No-show dataset, sourced from kaggle, was analyzed using Postgresql and visualized with Power BI. Patient No-show substantially affects the operational efficiency and resource allocation which leads to huge financial losses and adversely affecting patient health. This study uses the Patient Appointment data to explore the pattern of no show in respect to patient demographics, chronic disease profile, communication method, geographical factors, scholarship status and lead times. Addressing these factors can reduce financial losses, patient adherence and optimize resource utilization.

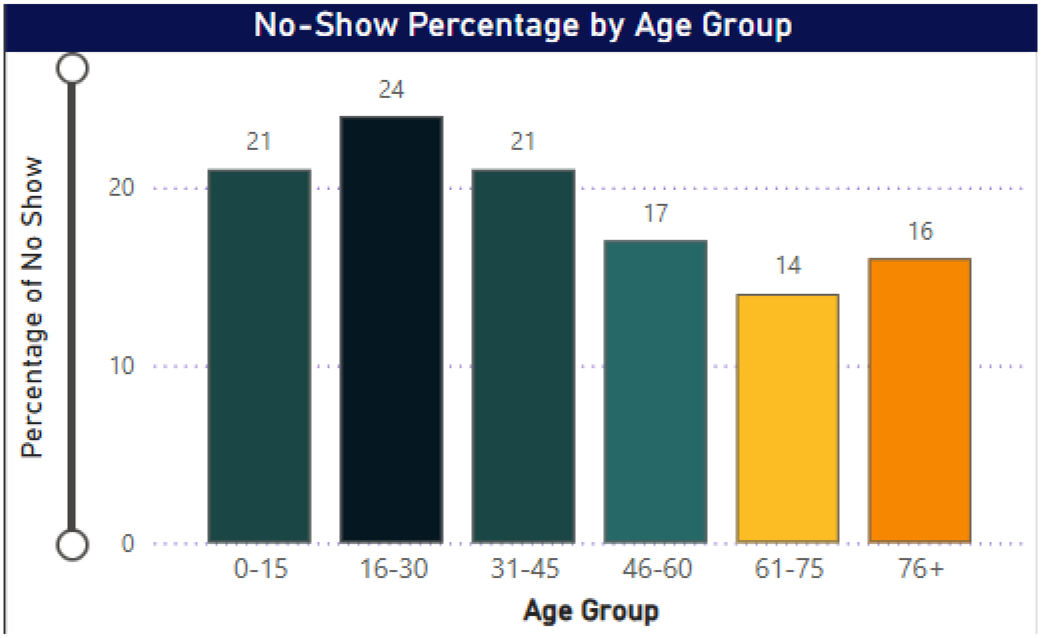
Key Findings

Chronic Diseases



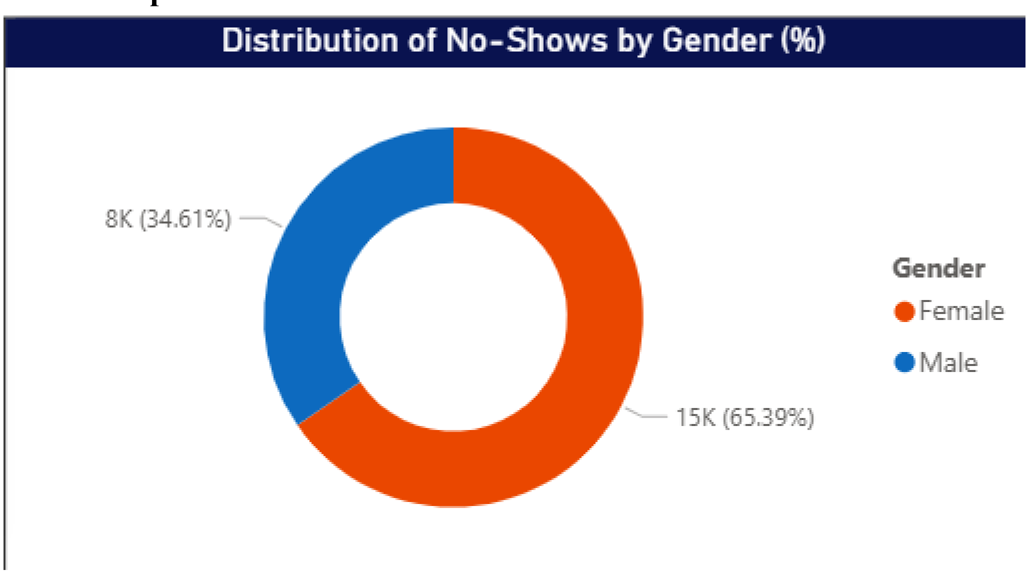
- Patient suffering from alcoholism accounts for the highest percentage of No shows in comparison to Number of Appointments is **17.67%** (with **420** of No shows of **2377** appointments) , in second place percentage of no show patient without chronic disease comes out to be **17.4%** (with **18k** of No shows of **103k** appointments).
- Surprisingly, patients with Hypertension & Diabetes showed slightly better appointment adherence.

Age Group Trends



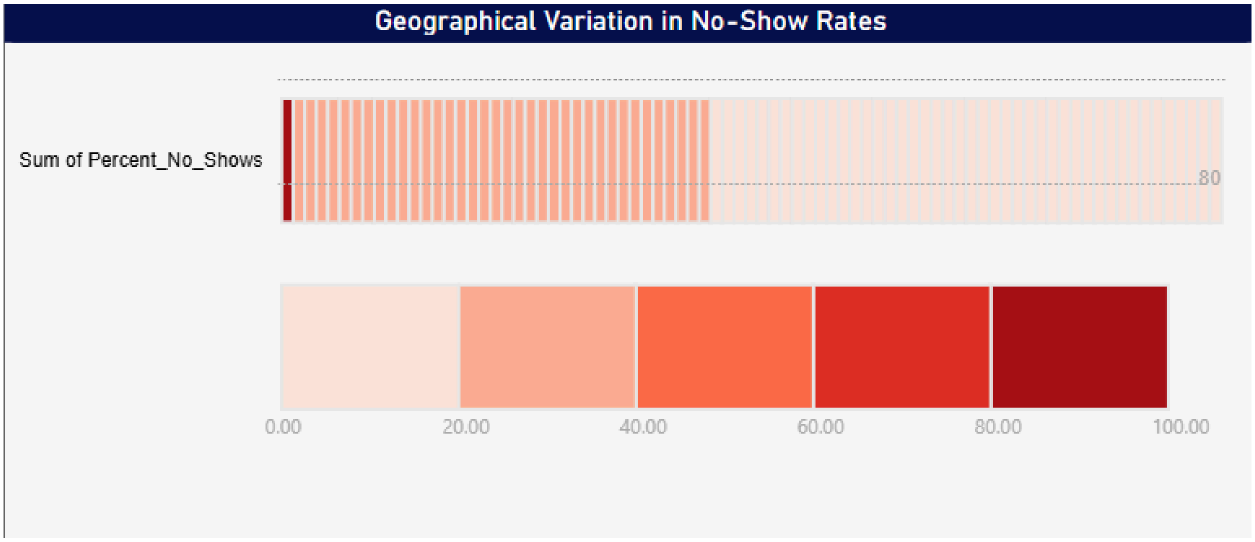
- No-show rates are highest in the **16–30 age group (24%)**, indicating a need for targeted engagement.
- Seniors (61–75, 76+ years) exhibit lower no-show rates, implying better health commitment.

Gender Impact



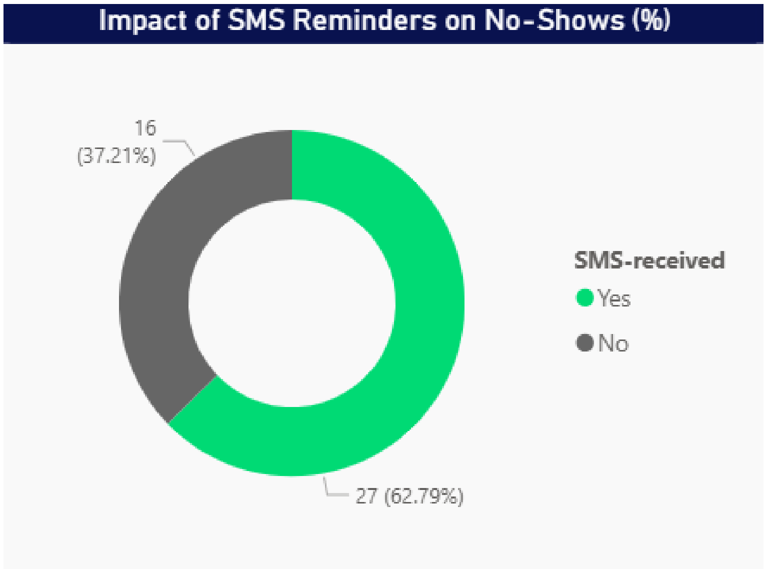
- **Females account for 65%** of total no-shows, suggesting gender-based behavioral patterns.

Geography



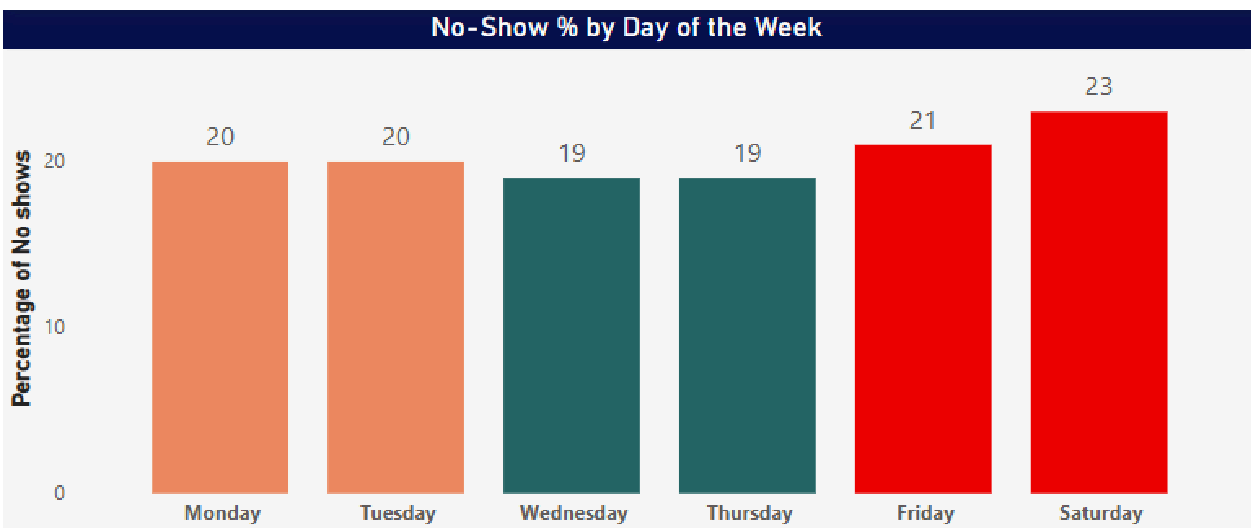
- Neighborhoods like **Santos Dumont (28%), Santa Cecilia (27%), and Santa Clara (26%)** exhibit the highest no-show rates, indicating localized barriers to appointment adherence, whereas areas like **Ilha do Boi (8%) and Solon Borges (14%)** show much better attendance.
- **Ilhas Oceanicas De Trindade shows 100%** of no-show rate but with only **2 appointments** and **Parque Industrial shows 0%** of no-show rate with only **1 appointment**.

Communication Effectiveness



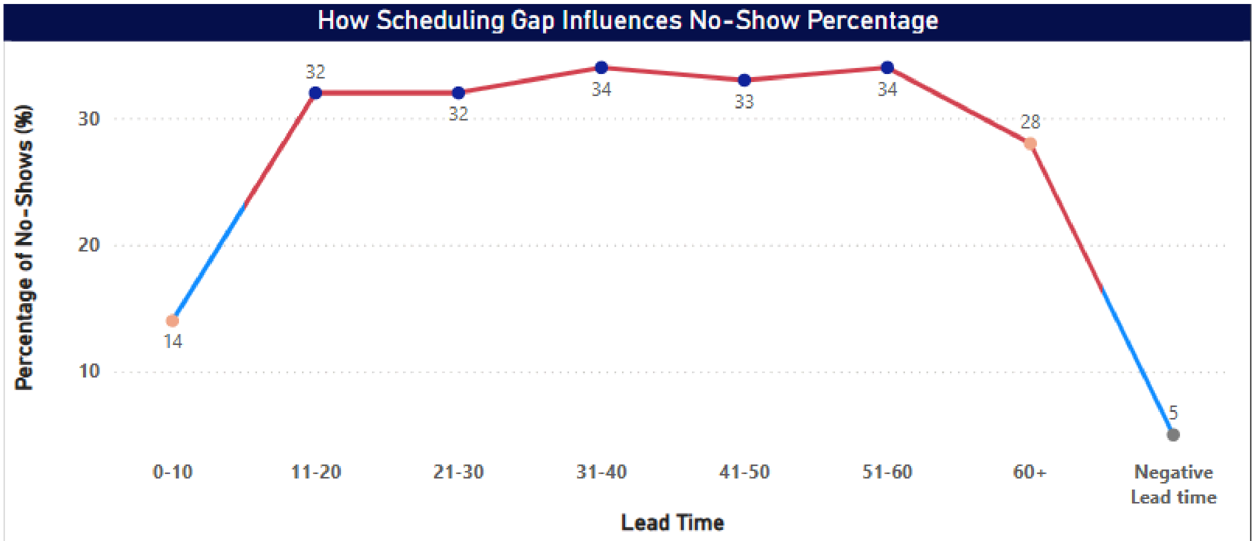
- Patients who received **SMS reminders showed significantly lower no-show rates (27% vs. 16%)**, validating the importance of proactive communication.

Weekday Patterns



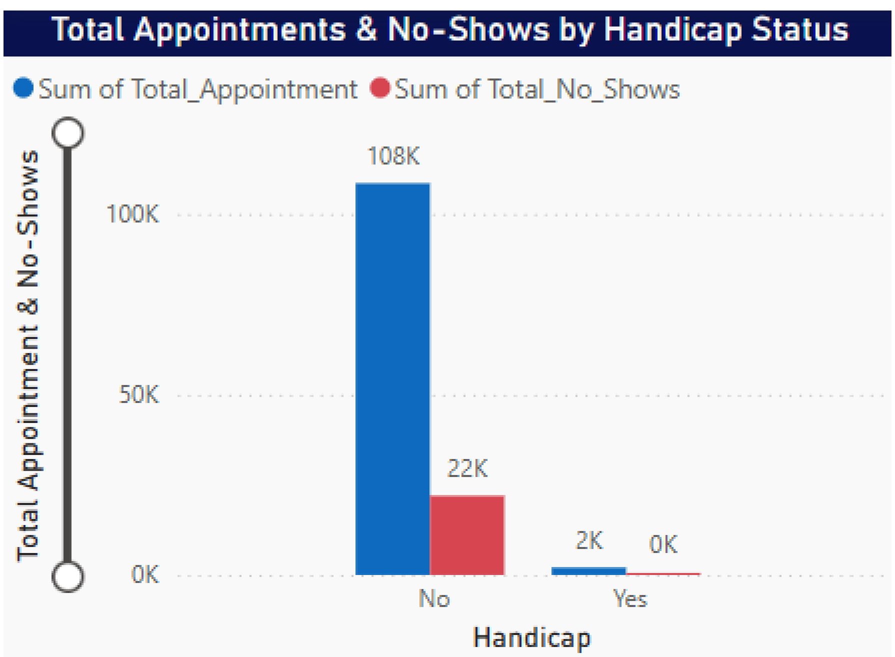
- Saturday sees the highest no-shows (23%), hinting at potential lifestyle factors affecting attendance.

Lead Time Influence



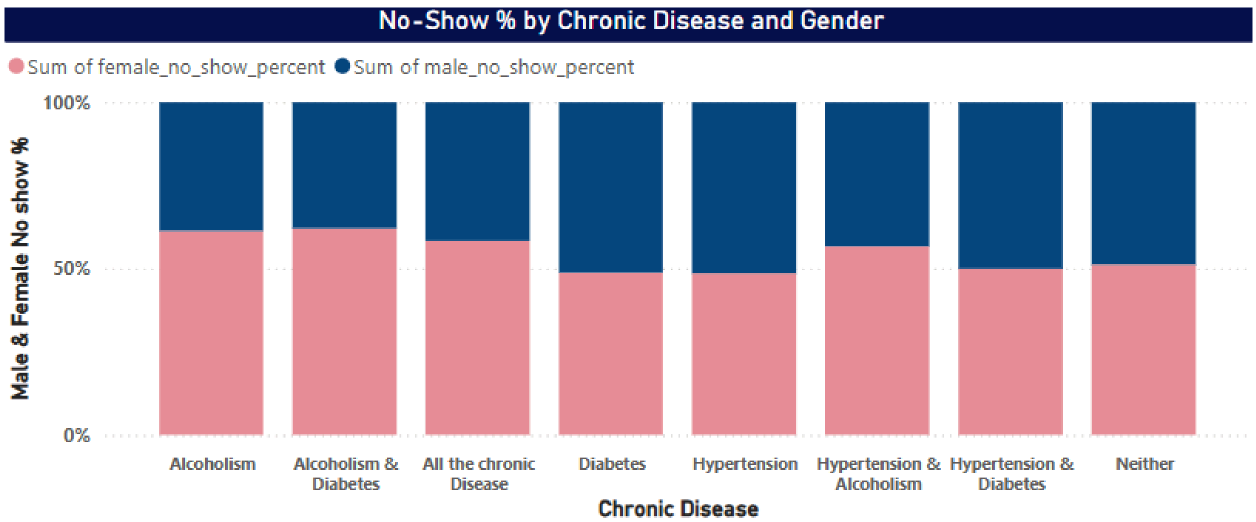
- Appointments scheduled **31–60 days in advance show the highest no-show rates (~33–34%)**, suggesting shorter lead times could improve adherence.
- Some data shows negative lead time which implies data problems.

Handicap Status



- Surprisingly, **non-handicapped patients missed more appointments** (approx~ 20%), indicating that disability status does not correlate with poor attendance (approx~ 17%).

Gender and Chronic disease status



- Females in general tend to miss their appointments more in comparison to Male.

Strategic Recommendations

- Lead time more than 10 days shows higher no show rates, Limiting the lead time would be the condition especially for young patients.
- Make appointments on Weekdays so reduce the no show rates as patients may want to enjoy their weekends.
- Increase SMS reminder to reduce the no-show rate.

Appendix

- https://www.kaggle.com/datasets/joniarroba/noshowappointments
- SQL Query Snippets
- Power BI Dashboard link~